CSE4088 Introduction to Machine Learning

Bias-variance Tradeoff

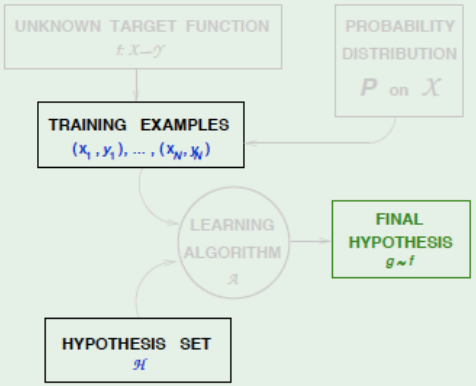
Slides are adopted from lecture notes of Yaser Abu-Mostafa

Review of last lecture

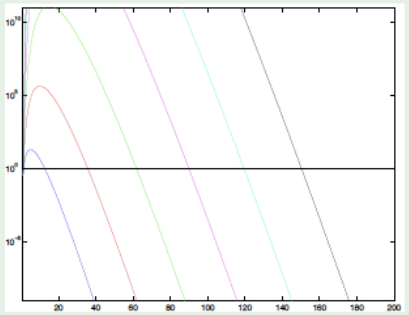
- VC dimension $d_{\text{VC}}(\mathcal{H})$

  most points $\mathcal{H}$ can shatter

- Scope of VC analysis



- Utility of VC dimension



$$N \propto d_{\text{VC}}$$

Rule of thumb: $N \geq 10\, d_{\text{VC}}$

- Generalization bound

$$E_{\text{out}} \leq E_{\text{in}} + \Omega$$

Ω is a function of N (number of samples), VC dimension and δ (prob. of error)

Outline

- Bias and variance
- Learning curves

Approximation-generalization tradeoff

Small E_{out}: good approximation of f out of sample.

More complex $\mathcal{H} \Longrightarrow$ better chance of **approximating** f

(You have to navigate through the hypothesis set using the data set you have. Your data set is the only source you have. You may end up with a hypothesis that is bad.)

Less complex $\mathcal{H} \Longrightarrow$ better chance of **generalizing** out of sample

Ideal $\mathcal{H} = \{f\}$ winning lottery ticket ☺

Quantifying the tradeoff

VC analysis was one approach: $E_{\text{out}} \leq E_{\text{in}} + \Omega$

Bias-variance analysis is another: decomposing E_{out} into

1. How well $\mathcal{H}$ can approximate f
2. How well we can zoom in on a good $h \in \mathcal{H}$

Applies to **real-valued targets** and uses **squared error**

Start with E_{out}

$$E_{\text{out}}(g^{(\mathcal{D})}) = \mathbb{E}_{\mathbf{x}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]$$

(E_{out} depends on the final hypothesis *g*, which depends on the dataset *D*)

(We want to release the dependency on the dataset by averaging out on D)

$$\mathbb{E}_{\mathcal{D}}\left[E_{\text{out}}(g^{(\mathcal{D})})\right] = \mathbb{E}_{\mathcal{D}}\left[\mathbb{E}_{\mathbf{x}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]\right]$$

(We can change the order of the expectations since the integrand is non-negative)

$$= \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]\right]$$

Now, let us focus on:

$$\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]$$

The average hypothesis

To evaluate $\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]$

we define the 'average' hypothesis $\bar{g}(\mathbf{x})$:

(This will give us the correct value, in theory, if the noise is zero-mean. This is not available in practice.)

$$\bar{g}(\mathbf{x}) = \mathbb{E}_{\mathcal{D}}\left[g^{(\mathcal{D})}(\mathbf{x})\right]$$

Imagine **many** data sets $\mathcal{D}_1, \mathcal{D}_2, \cdots, \mathcal{D}_K$

$$\bar{g}(\mathbf{x}) \approx \frac{1}{K}\sum_{k=1}^{K} g^{(\mathcal{D}_k)}(\mathbf{x})$$

Using $\bar{g}(\mathbf{x})$

$$\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right] = \mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}) + \bar{g}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]$$

$$= \mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x})\right)^2 + \left(\bar{g}(\mathbf{x}) - f(\mathbf{x})\right)^2 + 2\ \left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x})\right)\ \left(\bar{g}(\mathbf{x}) - f(\mathbf{x})\right)\right]$$

$$= \mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x})\right)^2\right] + \left(\bar{g}(\mathbf{x}) - f(\mathbf{x})\right)^2$$

Bias and variance

$$\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x})\right)^2\right]}_{\text{var}(\mathbf{x})} + \underbrace{\left(\bar{g}(\mathbf{x}) - f(\mathbf{x})\right)^2}_{\text{bias}(\mathbf{x})}$$

$$\text{Therefore,} \quad \mathbb{E}_{\mathcal{D}}\left[E_{\text{out}}(g^{(\mathcal{D})})\right] = \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]\right]$$

Var: How far is your hypothesis *g* from the average hypothesis (the «best» possible one)? It is the variance of the model due to the finite dataset.

Bias: How far is your «best hypothesis» from the target function? (You did your best by averaging out but it is still far from the target function)

$$= \mathbb{E}_{\mathbf{x}}[\text{bias}(\mathbf{x}) + \text{var}(\mathbf{x})]$$

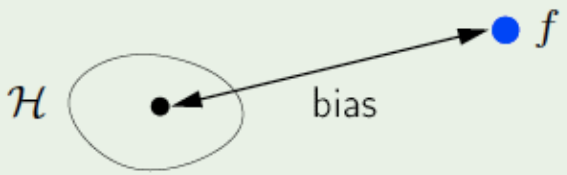
$$= \text{bias} + \text{var}$$

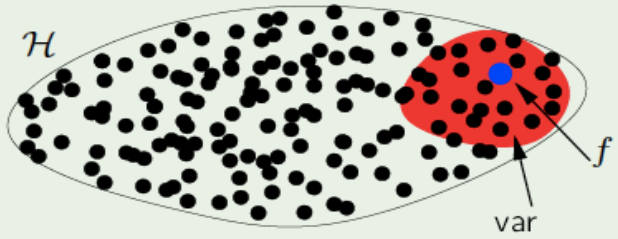
Interpretation: You learn from one dataset and calculate your error E_{out}. You do this for other possible datasets and take the average. Lets say your error is 0.3. Then, 0.05 of it might come from bias and 0.25 of it might come from variance.

The tradeoff (generalization vs. approximation)

$$\text{bias} = \mathbb{E}_{\mathbf{x}}\left[\left(\bar{g}(\mathbf{x}) - f(\mathbf{x})\right)^2\right] \qquad \text{var} = \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x})\right)^2\right]\right]$$

Small hypothesis set, bias is big





Huge hypothesis Set, includes f.

$\downarrow$ $\mathcal{H}\uparrow$ $\uparrow$

Bias diminishes, variance gets larger as the hypothesis set gets bigger since the red area gets bigger.

Each black point denotes a single hypothesis

Example: sine target

$f : [-1, 1] \to \mathbb{R}$ $\quad f(x) = \sin(\pi x)$

Only two training examples! $\quad N = 2$

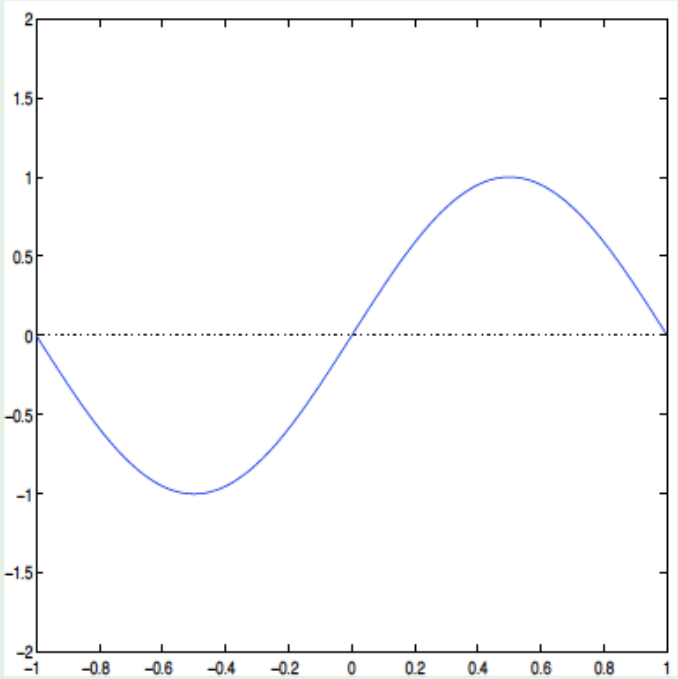
Two models used for learning:

$$\mathcal{H}_0: \quad h(x) = b$$

$$\mathcal{H}_1: \quad h(x) = ax + b$$

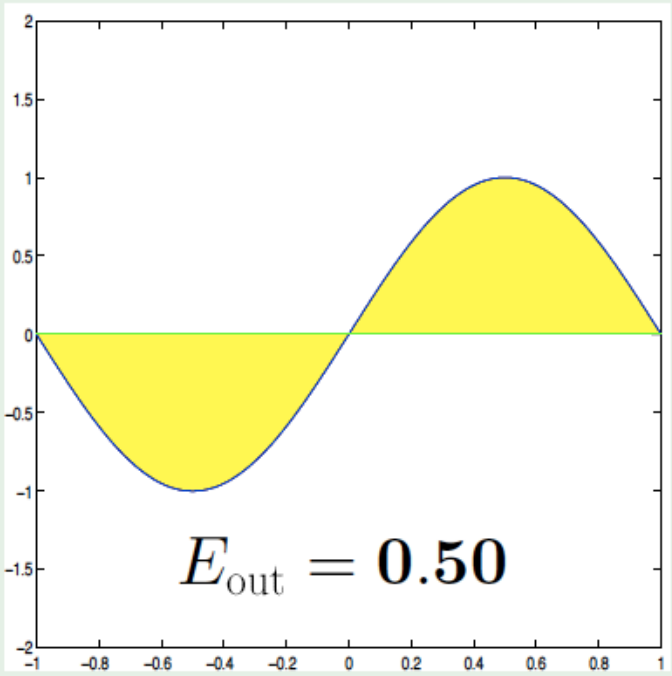
Which is better, $\mathcal{H}_0$ or $\mathcal{H}_1$?

f

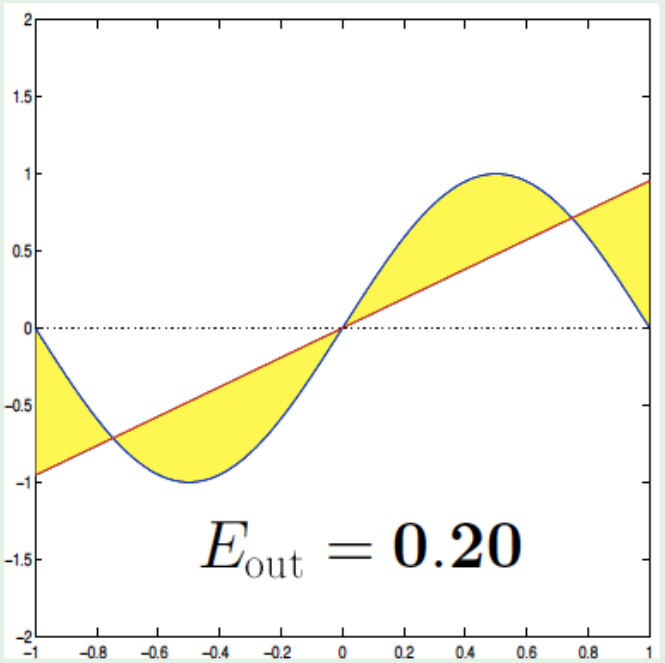


Approximation - $\mathcal{H}_0$ versus $\mathcal{H}_1$

$\mathcal{H}_0$



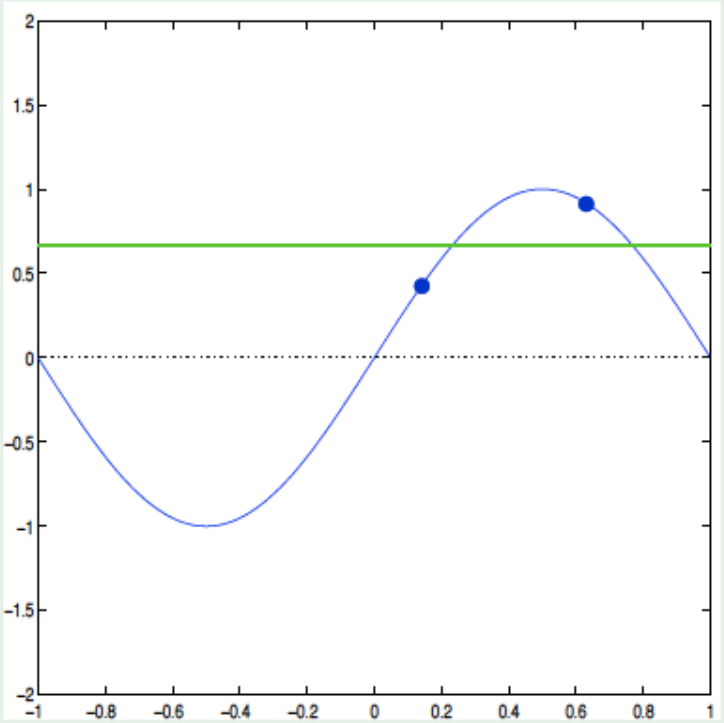
$\mathcal{H}_1$



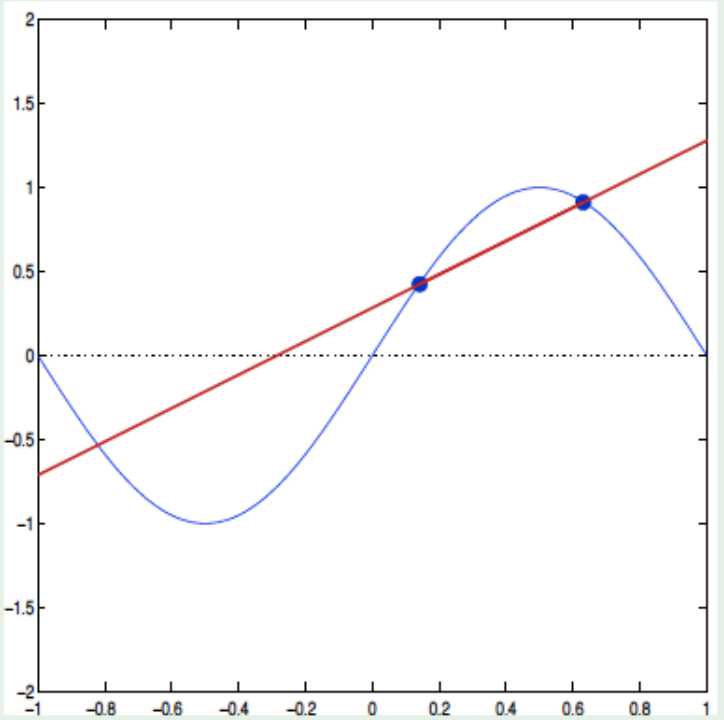
The linear model (more complex model) wins in terms of approximation since its error is smaller.

Learning - $\mathcal{H}_0$ versus $\mathcal{H}_1$

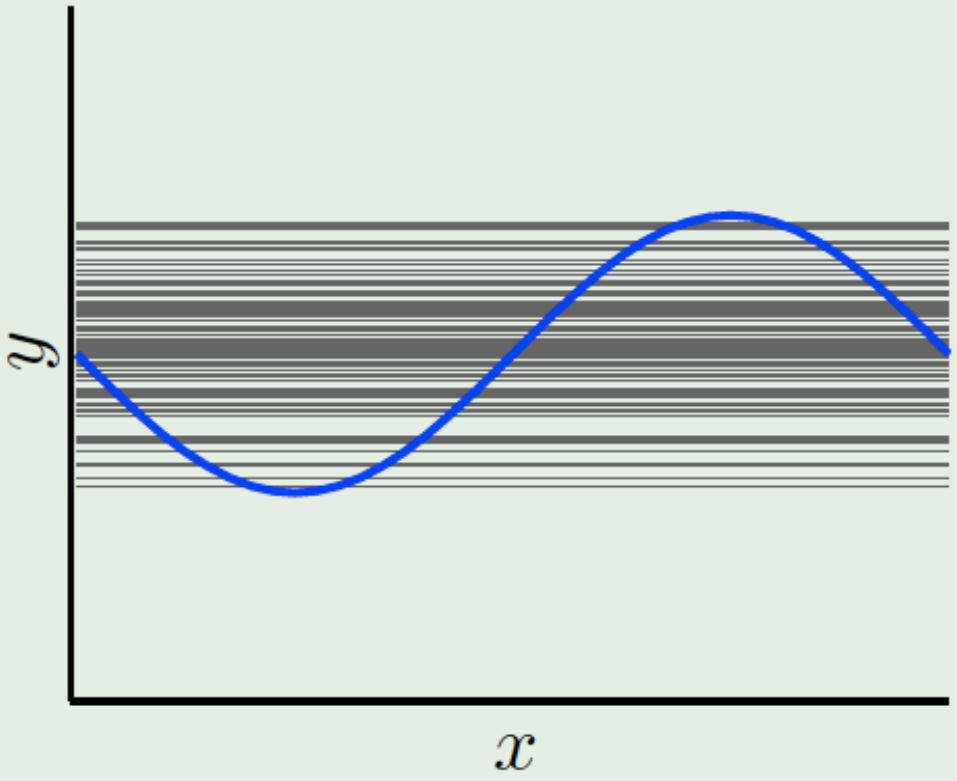
$\mathcal{H}_0$

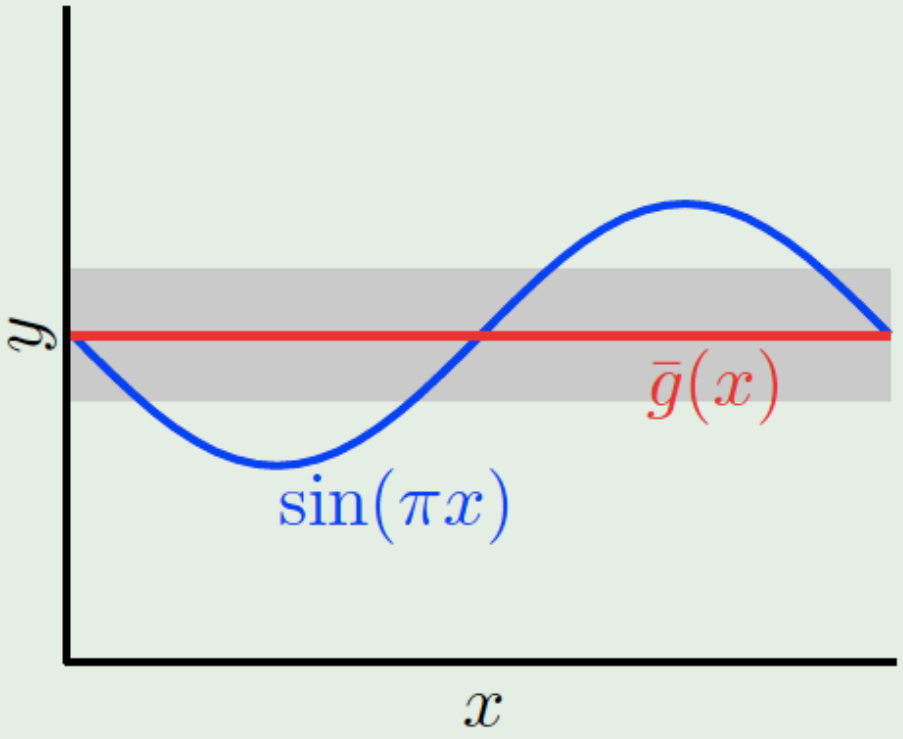


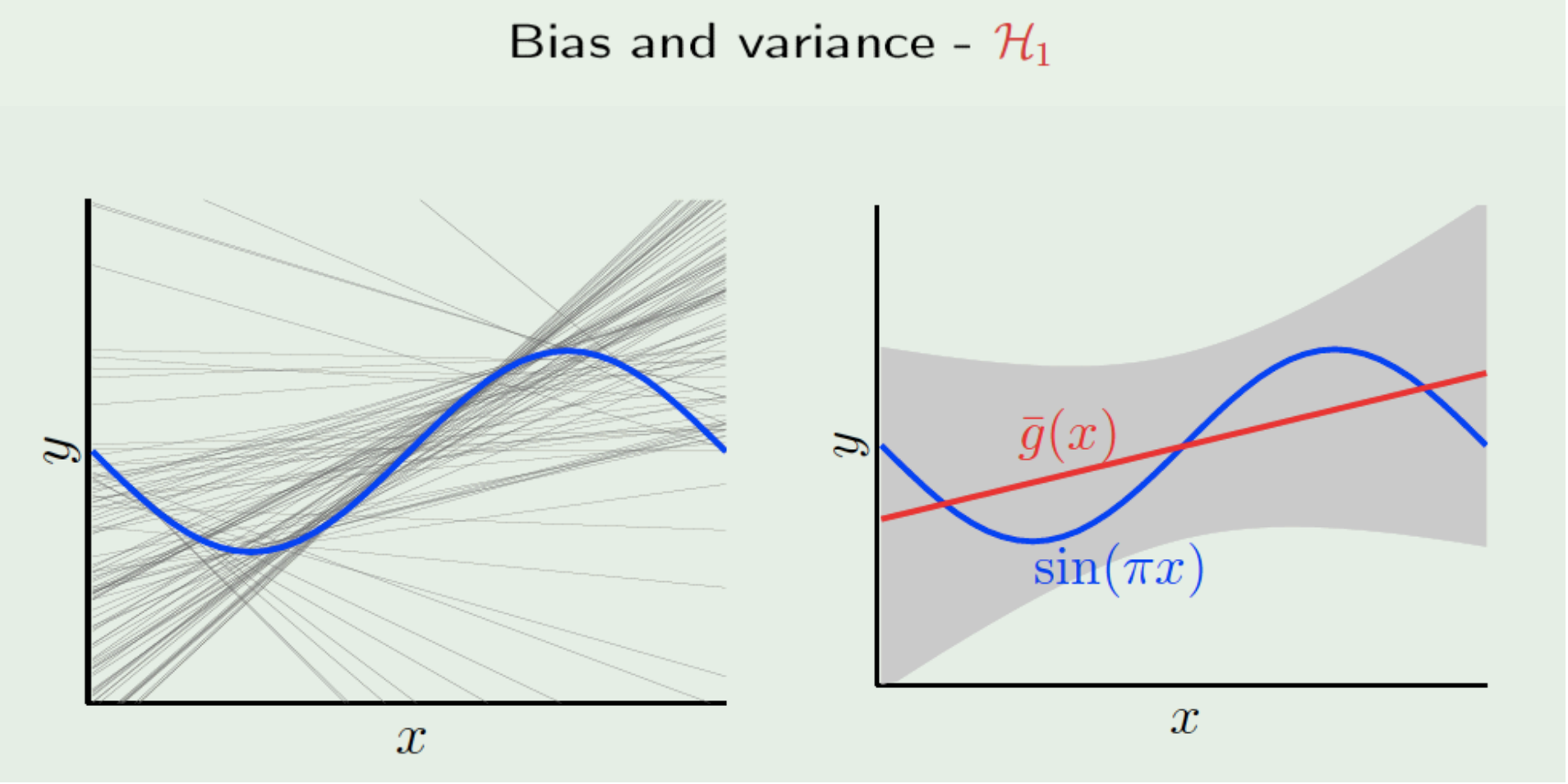
$\mathcal{H}_1$



Bias and variance - $\mathcal{H}_0$

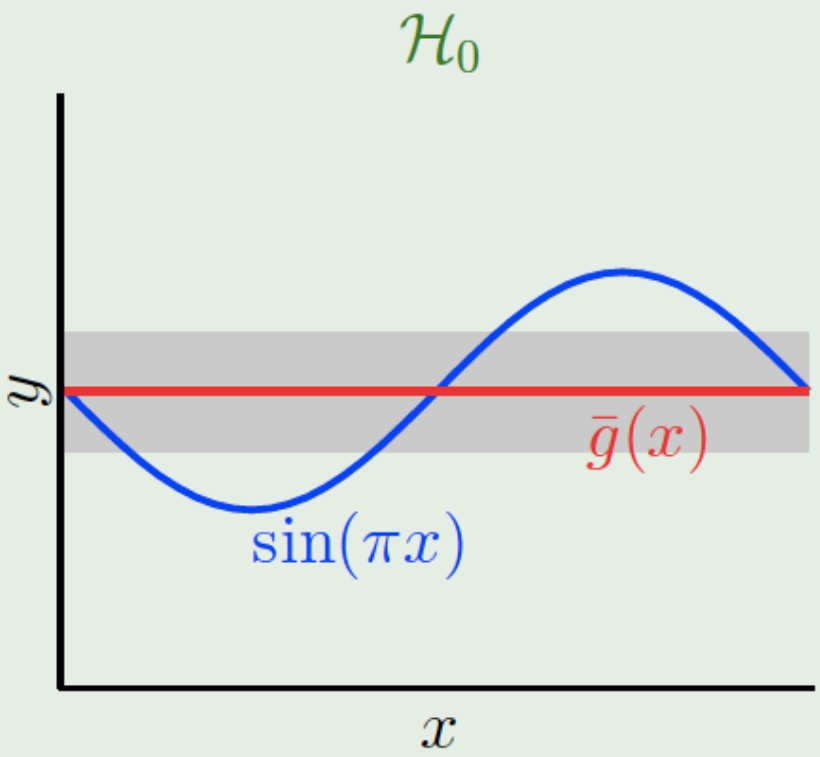




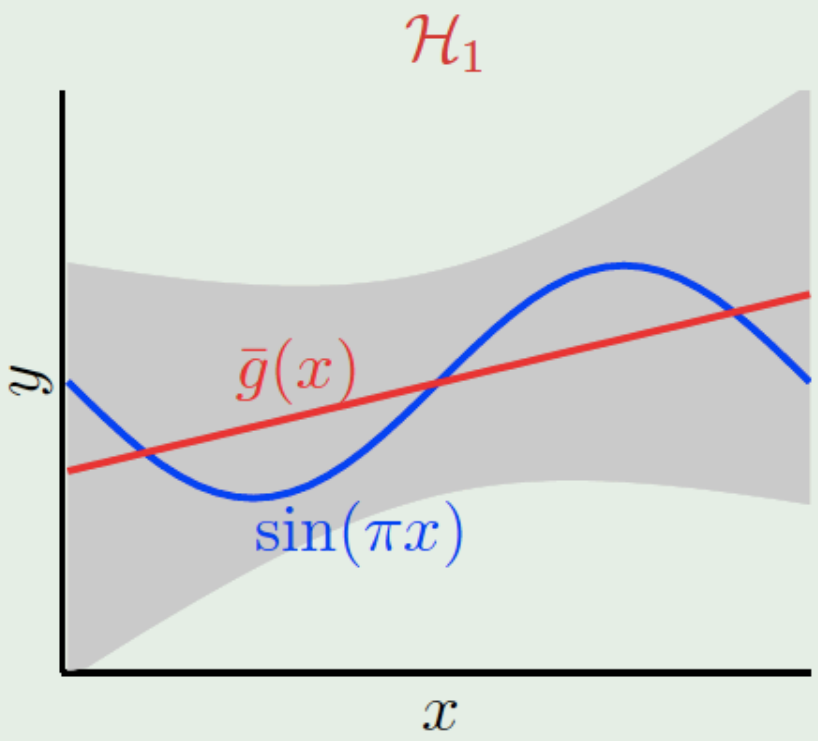


Question: From the point of learning: I have two points from a target function we do not know, is it better to use a constant or a line?

and the winner is ...



bias $= \mathbf{0.50}$ var $= \mathbf{0.25}$



bias $= \mathbf{0.21}$ var $= \mathbf{1.69}$

Error of (H_0) = bias + var = 0.50 + 0.25

Error of (H_1) = bias +var = 0.21 + 1.69 is larger

Lesson learned

Match the '**model complexity**'

to the **data resources**, not to the **target complexity**

Data resources mean: What do you have to navigate the hypothesis set? Choose the hypothesis set that you can afford to navigate with the data you have.

Expected E_{out} and E_{in}

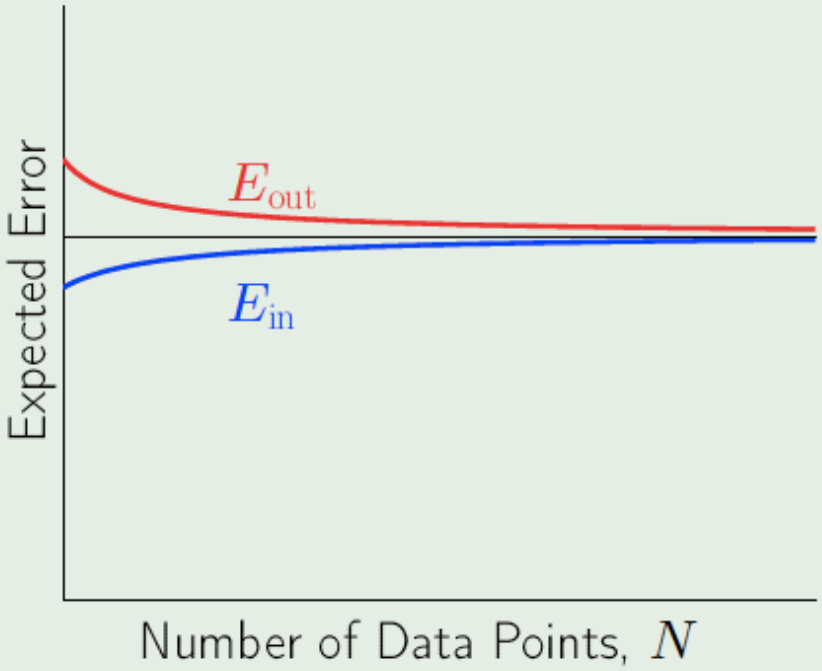
Data set $\mathcal{D}$ of size N

Expected out-of-sample error $\mathbb{E}_{\mathcal{D}}[E_{\text{out}}(g^{(\mathcal{D})})]$

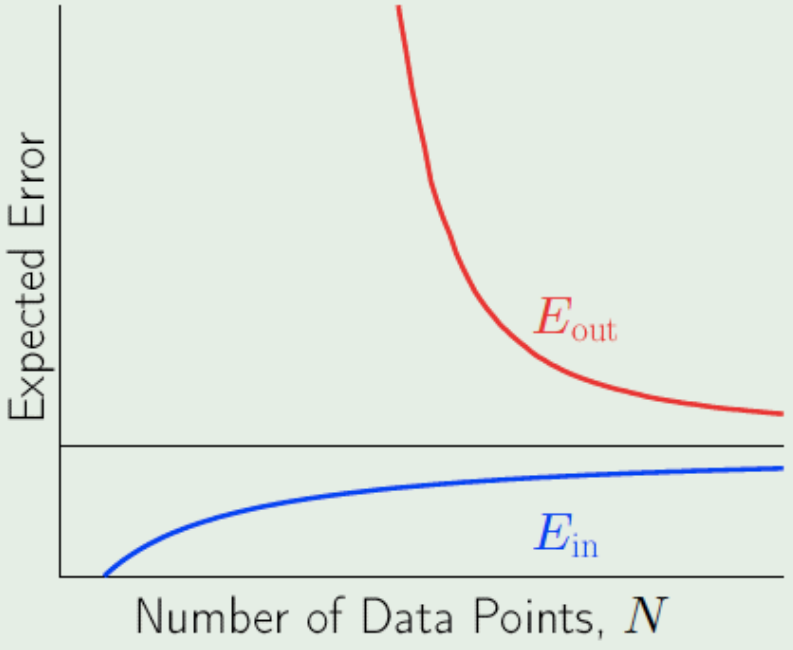
Expected in-sample error $\mathbb{E}_{\mathcal{D}}[E_{\text{in}}(g^{(\mathcal{D})})]$

How do they vary with N?

The curves

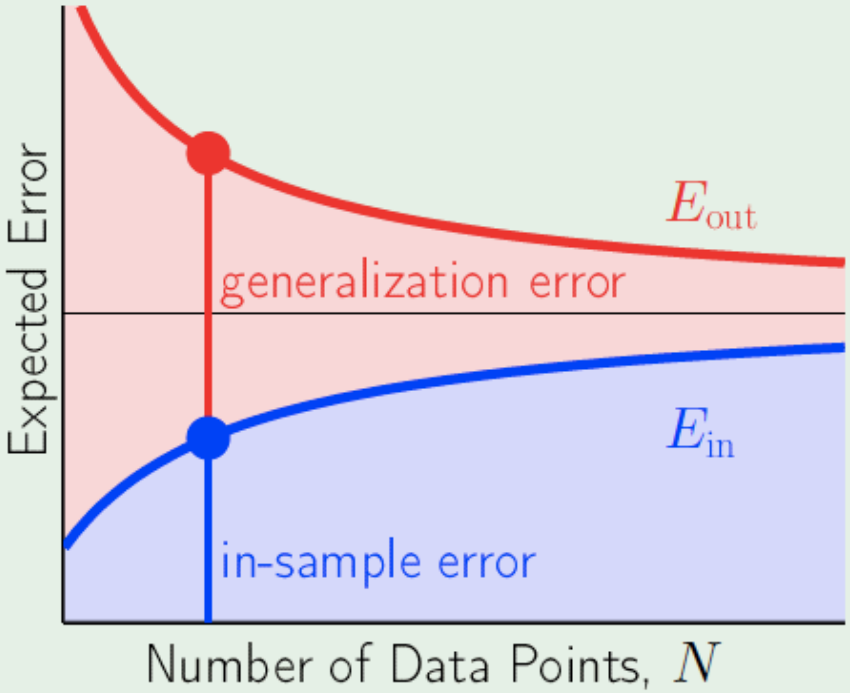


Simple Model

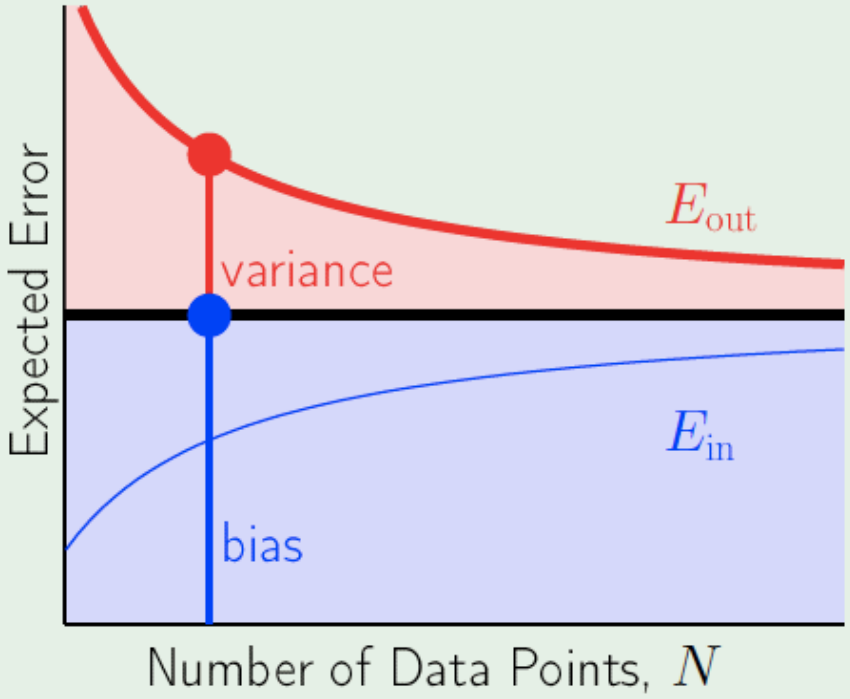


Complex Model

VC versus bias-variance



VC analysis



bias-variance

Linear regression case

Data set $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$

Noisy target: $y = \mathbf{w}^T\mathbf{x} + \text{noise}$

Linear regression solution: $\mathbf{w} = (\mathrm{X}^\mathsf{T}\mathrm{X})^{-1}\mathrm{X}^\mathsf{T}\mathbf{y}$

In-sample error vector $= \mathrm{X}\mathbf{w} - \mathbf{y}$

'Out-of-sample' error vector $= \mathrm{X}\mathbf{w} - \mathbf{y}'$

(Out-of-sample error will be obtained by using the same samples but different noise to generate y': this is in fact not allowed for computing E_{out}, we need to use samples not seen before).

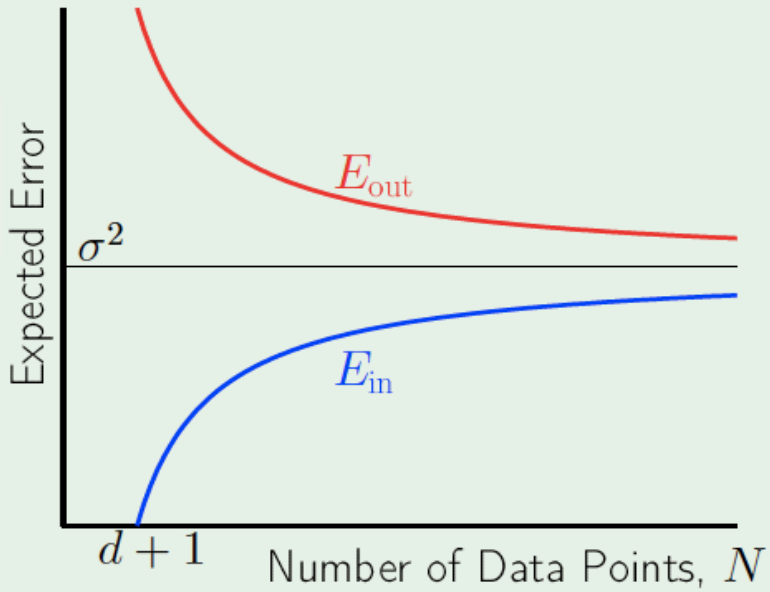
Learning curves for linear regression

Best approximation error $= \sigma^2$

Expected in-sample error $= \sigma^2 \left(1 - \frac{d+1}{N}\right)$

Expected out-of-sample error $= \sigma^2 \left(1 + \frac{d+1}{N}\right)$

Expected generalization error $= 2\sigma^2 \left(\frac{d+1}{N}\right)$



Sigma squared is the variance of the noise.
d is the dimensionality of the data.